

Name _____ Date _____

10.3 Logic

Getting started

Complete the following sentences:

The _____ gate only returns 1 when both inputs are 1.

The _____ gate only returns 0 when both inputs are 0.

The _____ gate only returns 1 if the input is 0.

The _____ gate returns 0 if both inputs are 0, and when both inputs are 1.

The _____ gate only returns 1 if both inputs are 0.

Practice

Correct the error(s) in the truth tables.

1 A AND (B OR C)

A	B	C	A AND (B OR C)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

2 A XOR NOT B

A	B	C	A XOR NOT B
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

3 (A NOR B) OR (C AND A)

A	B	C	(A NOR B) OR (C AND A)
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Challenge

You only need to work with a logic diagram, or expression, that has one output. They can, however, have more than one output.

Draw a logic diagram for the following 2-output statements.

1 $X = A \text{ AND } B$

$$Y = C \text{ OR } D$$

2 $P = \text{NOT}(A \text{ AND } B)$

$$Q = (A \text{ XOR } B) \text{ OR } C$$

3 $S = (A \text{ NOR } B) \text{ AND } C$

$$T = \text{NOT}(A \text{ OR } B)$$